Independent Computational Verification

Paper II — Section 6: the complete-split cover program (Prop 6.2, Cor 6.3, Lemma 6.1)

Author of record: Juan Pablo Traverso Gianini **Date:** 2026-07-09
Artifact: `D_completesplit_cover/audit_D_completesplit.py` **Verdict: PASS**

Scope

All complete-split graphs $S_{p,q} = K_p \vee \overline{K_q}$ with $p \le 12$, $q \le 14$, $p + q \le 16$ — **139 cases**, including the degenerate $S_{2,0}$ and $p \le 1$.

Claims verified

- **(D1) Prop 6.2:** the full-graph LP value $\tau_3^*(S_{p,q})$ equals the closed form 0 (if $p \le 1$ or $(p, q) = (2, 0)$), $\frac{\binom{p}{2}+pq}{3}$ (non-saturated $q \le p - 1$, $p \ge 3$), or $\binom{p}{2}$ (saturated $q \ge p - 1$).
- **(D2) Cor 6.3:** $\Phi_\tau(S_{p,q}) = |E| - 2\tau_3^*$ equals $pq - \binom{p}{2}$ (saturated) or $\frac{\binom{p}{2}+pq}{3}$ (non-saturated).
- **(D3) Lemma 6.1:** the orbit-averaged 2-variable LP attains the **same** optimum as the full LP.

Results (sample; full set in `results/summary.json`)

p	q	τ_3^* formula	τ_3^* LP	Φ_τ formula
0	7	0.0	0.0	0.0
0	8	0.0	0.0	0.0
0	9	0.0	0.0	0.0
0	10	0.0	0.0	0.0
0	11	0.0	0.0	0.0
0	12	0.0	0.0	0.0
0	13	0.0	0.0	0.0
0	14	0.0	0.0	0.0
1	0	0.0	0.0	0.0
1	1	0.0	0.0	1.0
1	2	0.0	0.0	2.0
1	3	0.0	0.0	3.0

- D1 (LP = formula) violations: **0**.
- D2 (Φ_τ formula) violations: **0**.
- D3 (orbit-LP = full LP) violations: **0**.
- **All 139 cases exact** (agreement to LP tolerance 10^{-7}). Runtime 0.58 s.

Verdict

PASS. The piecewise cover formula, the Φ_τ values, and the sufficiency of orbit averaging are confirmed exactly across the tested range.

This document reports an ***independent computational verification*** *of the stated claims: each result was recomputed from scratch by a self-contained script (exact linear programming via SciPy/HiGHS for the fractional triangle-cover number, exact rational arithmetic for the integer optimization, and exhaustive enumeration of non-isomorphic graphs via the NetworkX atlas where indicated). This is computational evidence corroborating the analytic proof; it is not itself a formal proof. The machine-checked formal proof is the Lean development, which is conditional on the classical chordal-structure theory.*